RoPAWS: Robust Semi-supervised Representation Learning from Uncrated Data

Sangwoo Mo[1], Jong-Chyi Su[2], Chih-Yao Ma[3], Mahmoud Assran[3,4,5], Ishan Misra[3], Licheng Yu[3], Sean Bell[3]

[1]KAIST, [2]NEC Laboratories America, [3]Meta AI, [4]McGill University, [5]MILA

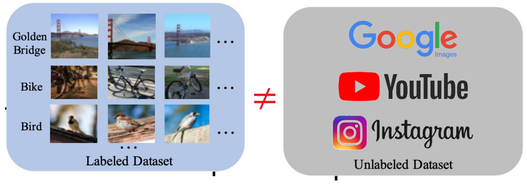
Paper: https://arxiv.org/abs/2302.14483

Code: https://github.com/facebookresearch/suncet

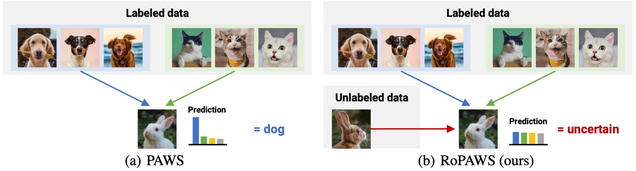
TLDR. Robust semi-supervised learning method derived from a novel probabilistic view of learned representations

Introduction

Goal. Semi-supervised learning (Semi-SL) where unlabeled data is uncurated
- Semi-SL learns a model using a small labeled data and large unlabeled data
- In real world, unlabeled data is often uncurated, e.g., collected from Google



Motivation. PAWS, the SOTA Semi-SL method, often fails in this uncurated scenario, because PAWS is overconfident in some class, and prone to the out-of-class data



Contribution. Fix PAWS to be robust on uncurated data (so called RoPAWS)
- Reinterpret PAWS as a generative classifier using KDE on embedding space
- Design a novel method using the probabilistic interpretation of PAWS
- Achieve SOTA on various (including large-scale) robust Semi-SL benchmarks

PAWS: The SOTA method for curated Semi-SL

Prediction. Apply soft nearest-neighbor (Soft-NN) classifier
- For a data x with embedding z, assign label y of the nearest labeled data z_l
- Soft version is a weighted average of labels over all labeled data $\mathcal{B}_l$

$$p_{\text{PAWS}}(y|x) := \sigma_\tau(z \cdot \mathbf{z}_l^\intercal) \cdot \mathbf{p}_l^{y|x}$$

Training. Make predictions of different views be identical (joint embedding method)

RoPAWS: Robust PAWS for uncurated Semi-SL

Probabilistic view of PAWS. Define marginal $p(x)$ and conditional $p(x|y)$ probabilties over embeddings using kernel density estimation (KDE):
- Here, KDE is defined only over the labeled data $\mathcal{B}_l$

$$p_{\text{PAWS}}(x|y) := \frac{1}{|\mathcal{B}_l^y|} \sum_{x' \in \mathcal{B}_l^y} k(z, z') \quad \text{and} \quad p_{\text{PAWS}}(x) := \frac{1}{|\mathcal{B}_l|} \sum_{x' \in \mathcal{B}_l} k(z, z')$$

Bayes' rule gives the prediction $p(y|x)$ recovering the original definition of PAWS

$$p_{\text{PAWS}}(y|x) = \frac{p_{\text{PAWS}}(x|y)\, p(y)}{p_{\text{PAWS}}(x)} = \frac{\sum_{\mathcal{B}_l^y} k(z, z')}{\sum_{\mathcal{B}_l} k(z, z')} = \sigma_\tau(z \cdot \mathbf{z}_l^\intercal) \cdot \mathbf{p}_l^{y|x}$$

Calibrated prediction. Extend the definition of $p(x)$ and $p(x)$ to consider both labeled data $\mathcal{B}_l$ and unlabeled data $\mathcal{B}_u$ (so that not be overconfident)

$$p_{\text{RoPAWS}}(x|y) = \frac{1}{K'} \sum_{\mathcal{B}_l \cup \mathcal{B}_u} p(y|x')\, k(z, z') \quad \text{and} \quad p_{\text{RoPAWS}}(x) = \frac{1}{K'} \sum_{\mathcal{B}_l \cup \mathcal{B}_u} \frac{1}{C}\, k(z, z').$$

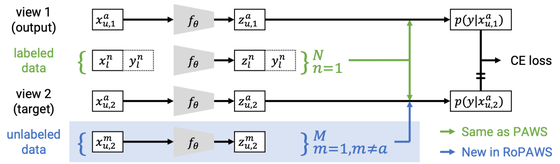
Bayes' rule gives the prediction $p(y|x)$ follows some closed-form solution
- It is not directly computed since $p(x|y)$ needs $p(y|x)$ of unlabeled data

$$\mathbf{p}_u^{y|x} = (\mathbf{I} - \mathbf{s}_u \odot C \boxed{\mathbf{p}_u^y})^{-1} \cdot (\mathbf{s}_l \odot C \boxed{\mathbf{p}_u^y} \cdot \boxed{\mathbf{p}_l^{y|x}})$$

Assumed prior — Assumed prior — Known for labeled data

Omitted contents. Check the paper for additional novelties, including:
- RoPAWS is a (principled) instantiation of label propagation
- Data-dependent in-domain prior to explicitly model out-of-class

Overall framework. Same as PAWS, but use the calibrated prediction as targets



Experiments

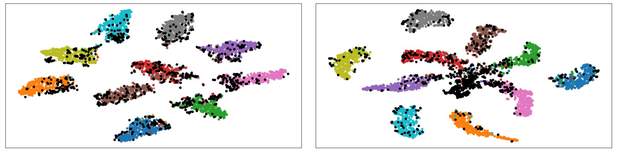
CIFAR-10 results. Achieves SOTA on various robust Semi-SL benchmarks

In-class	CIFAR-10		CIFAR-100	
Out-of-class	SVHN	Tiny-IN	SVHN	Tiny-IN
FixMatch (Sohn et al., 2020)	68.0±0.7	70.5±1.2	41.7±1.3	46.0±1.0
PAWS (Assran et al., 2021)	47.1±0.4	60.4±5.3	36.9±0.2	46.0±0.5
UASD (Chen et al., 2020d)	66.7±1.0	74.0±0.4	39.5±0.8	44.6±0.8
RealMix (Nair et al., 2019)	58.2±5.3	69.2±2.3	44.1±1.0	47.6±1.4
DS^{3}L (Guo et al., 2020)	50.0±8.3	69.1±2.3	29.0±1.0	40.2±0.9
OpenMatch (Saito et al., 2021)	74.7±1.0	78.0±0.8	29.0±1.3	32.2±2.6
OpenCoS (Park et al., 2021)	79.0±1.1	82.5±1.3	49.2±1.2	54.0±1.8
RoPAWS (ours)	**81.2**±0.4	**83.1**±1.8	**51.5**±1.7	**59.8**±1.3

Semi-iNat results. Also scalable to large-scale benchmark. Some remarks:
- Learning from large uncurated data (superset) > small curated data (subset)
- Also effective for transfer learning scenario (use ImageNet pre-trained model)
- Slightly helps (not harms) the curated scenario using calibrated prediction

	From scratch		ImageNet pre-trained	
Method	Curated	Uncurated	Curated	Uncurated
Pseudo-label	18.6	18.8	40.3	40.3
FixMatch	15.5	11.0	44.1	38.5
Self-training	20.3	19.7	42.4	41.5
FixMatch + CCSSL	-	31.2	-	-
MoCo-v2 + Fine-tuning	30.2	31.8	41.7	40.8
MoCo-v2 + Self-training	32.0	32.9	42.6	41.5
MoCo-v2 + FixMatch + CCSSL	-	41.3	-	-
PAWS	39.2	37.7	49.7	43.0
RoPAWS (ours)	39.4 (+0.2)	43.0 (+5.3)	50.3 (+0.6)	44.3 (+1.3)

t-SNE visualization. Well separates out-of-class (black) and in-class (colored) data



(a) PAWS (b) RoPAWS (ours)